Voxel Camera with Laser Tracking

REQUIREMENTS AND GATE 1

Vocel camera

Milestone M2

6 November 2026

GATE 1: PENDING

Team approval has not yet been confirmed.

Approval date: not yet recorded.

System in a few sentences

This laboratory demonstrator tracks a ball swinging on a string inside a cube and keeps a galvo-steered laser spot on it. Three camera nodes observe the ball and the spot; their observations support a 3D position estimate, prediction and aiming-error evaluation (F-VLT-1 to F-VLT-7).

The field of application is a distributed real-time system with embedded sensing and actuation. The physical loop is ball and laser spot -> cameras -> networked estimation and prediction -> galvo and laser -> observed ball and spot. Correct operation depends on both spatial accuracy and timely data (NF-VLT-1, NF-VLT-5, NF-VLT-18, F-VLT-22).

Scope of this report

Selected requirements cover the core physical loop, timing, fault handling, non-interference and verification. IDs retain the project numbering. Detailed node architecture, message layouts, module designs and transaction budgets belong to the next milestone, after Gate 1. Existing hardware and operating constraints are recorded as decisions, not newly designed here.

This is a self-contained review draft, not an approval record. Open decisions and their owners are listed at the end. No individual approver names are recorded.

Core function and success criterion

Selected functional requirements and their measurable outcomes. Spatial thresholds are project targets, not values supplied by the lecture.

F-VLT-1 | Ball observation

Each camera reports the ball image position, or not seen, for every captured frame, with its exposure timestamp in ns of the shared clock. Reason: Fusion needs a measurement time rather than just a packet arrival time.

Acceptance: In at least 95% of frames with an annotated ball, detect it within 3 px of the label; no-ball frames report not seen.

Timing basis: [C] global time / RT images; [L] p. 3. Timestamp semantics remain OQ-12.

F-VLT-2 | Spot observation

Each camera reports the spot position, or not seen, with the ball in the same observation (exposure time in ns). Reason: The physical aiming result must be observed.

Acceptance: At least 95% of annotated visible spots are detected within 3 px; at least 99% of laser-off frames report no spot.

Timing basis: [C] RT images / monitored actuator effect; [L] p. 3.

F-VLT-3 | 3D ball position

Compute cube-coordinate ball position from at least two calibrated cameras, matched by exposure time (ns). Reason: Tracking must retain depth information when one camera is lost.

Acceptance: At 10 or more hand-measured resting positions spread across the cube, error is at most 10 mm with three cameras and 15 mm with every two-camera pair.

Timing basis: [C] timestamp comparison / temporal accuracy; [L] p. 3; limits OQ-12.

F-VLT-4 | Prediction at placement time

Aim at the ball position predicted for the placement timestamp (ns), not the last measured position. Reason: The ball moves during sensing, communication and mirror motion.

Acceptance: Replay of recorded swings gives a median prediction error of at most 10 mm against independent position references at the placement times. Reference method remains OQ-9.

Timing basis: [C] temporal accuracy / action delay; [L] pp. 3, 5, 11; NF-VLT-18.

F-VLT-5 | Physical aiming

Convert a target point into galvo output using the stored manual calibration. Reason: An accurate estimate alone does not establish an accurate physical aim.

Acceptance: For nine known reference points, the observed spot is within 5 mm of each point. The physical target/measurement procedure remains OQ-13.

F-VLT-6 | Measured aiming error

Compute aiming error when the observed geometry permits measurement; the existing single-camera rule needs clarification. Reason: An output command is not proof that the spot reached the ball.

Acceptance: Reported error is within 5 mm of an independently annotated reference. OQ-9 must settle the metric, single-camera validity and unmeasurable cases.

F-VLT-7 | Run evaluation

Detect release and run end, count hits and report hit rate. Current amplitude thresholds are 100 mm to start and below 50 mm to end; the hit radius is 25 mm. Reason: The demo needs repeatable evaluation, including missed observations.

Acceptance: Run boundaries are detected within one measured swing period T (s) of their annotations; hit rate matches the reference within 1 percentage point. Release, gap and time-weighting rules remain OQ-10.

Timing basis: [C] temporal accuracy; [L] p. 3. The swing-period duration in s is to be measured (OQ-10).

NF-VLT-1 | Demo success

Achieve at least 80% hit rate from release until amplitude falls below 50 mm. Reason: This expresses the primary observable purpose of the demonstration.

Acceptance: Three consecutive runs each achieve at least 80% using the scoring rule of F-VLT-7. OQ-9/OQ-10 and degraded-mode scope in OQ-14 must be settled first.

Valid data and recovery

The following requirements prevent delayed, inconsistent or pre-restart data from becoming current control input.

F-VLT-22 | Temporal validity

Bound observation age, exposure skew, prediction horizon, target age and sync-status age in ms; express target expiry in shared-clock ns. New packets based on expired images are not fresh. A clock correction must not extend the target-loss timeout. Reason: Shared timestamps alone do not guarantee timely physical data.

Acceptance: Inject inputs immediately inside/outside each configured validity bound, plus old-source targets and clock steps: expired inputs never drive the laser; local target-loss timing is not extended. Numerical bounds remain OQ-12.

Timing basis: [C] temporal accuracy, timestamp uncertainty, permanence/action delay; [L] pp. 3, 5, 11.

F-VLT-23 | Message validity and bounded buffering

Use a shared versioned contract; validate sender, structure, finite values, restart session and sequence order including wrap. Invalid, duplicate and obsolete-session messages must not refresh state or timeouts. Buffering is finite. Reason: Independently developed nodes must agree on what data is current.

Acceptance: Inject malformed messages, wrong versions/IDs, reordering, duplicates, counter wrap, restart and bursts beyond capacity: only current valid messages update state; memory remains bounded and expired data is discarded.

Timing basis: [C] action delay / hidden channels / peak load; [L] pp. 5, 9, 11. Queue limits remain OQ-12.

F-VLT-24 | Configuration and calibration consistency

Validate units, coordinate convention, camera IDs/image modes, setup revision, finite calibration parameters and cross-field configuration limits before arming. Reject incompatible peer data. Reason: Wrong calibration can look like a valid measurement while producing a wrong aim.

Acceptance: Remove/corrupt calibration, swap camera assignments, change image mode or setup revision, and set invalid limits: reject the affected setup/data and keep the laser off if valid prerequisites are absent. Report a specific diagnostic.

F-VLT-17 | Camera reintegration

A restarted camera can rejoin without restarting the other nodes, after time synchronization and setup validation. Reason: A single recoverable camera outage must not require a full-system restart.

Acceptance: Restart a camera while the remaining pair tracks: old-session data is rejected; validated observations can restore normal mode. Ground state, retained calibration/configuration and the reintegration/re-arm point are defined before implementation (OQ-11).

Timing basis: [C] reintegration / minimal history state / membership; [L] p. 4. No fixed rejoin duration in s is claimed.

Verification basis for these requirements

Use a time-synchronized, annotated reference set from the intended setup: at least 10 resting positions with measured 3D coordinates; swinging runs; visible and missing spots; laser-off and no-ball frames. Preserve camera calibration and exposure timestamps in ns. Do not record people. Re-record after relevant setup changes (F-VLT-1 to F-VLT-7, F-VLT-24).

Replay the same processing logic with a simulated clock, controlled message delivery and captured actuator output. Use repeated schedules to check deterministic validity/state decisions, then verify physical aiming and timing on the real installation. Independent moving-position/error references remain OQ-9 (F-VLT-4, F-VLT-6, F-VLT-22, F-VLT-23).

Laser-off behaviour and fault tolerance

Software protections are specified separately from the independent physical supply interruption. Recovery policy is still open in OQ-11.

F-VLT-8 | Off by default

Keep the laser off at startup, while unarmed and when no valid ball state exists. The current all-node-restart rule must be reconciled with two-camera operation. Reason: Actuation requires valid prerequisites.

Acceptance: Power-up, disarming and loss of a valid state show laser-off within 100 ms. Startup must not produce an on pulse. Camera-restart expectations remain OQ-11.

Time value: project target. Timing basis: [C] fault hypothesis / actuator effect; [L] pp. 4, 5, 10.

F-VLT-9 | Bounded aiming envelope

Constrain every commanded aim to the cube angle envelope and reject non-finite targets with the laser off. Reason: Received coordinates must not bypass local actuator limits.

Acceptance: Outside-volume, far-out-of-range, NaN and infinity targets never command angles outside the calibrated envelope; non-finite targets leave the laser off. Envelope dimensions remain OQ-3.

F-VLT-10 | Loss-of-target shutdown

Switch the laser off after 100 ms without a fresh accepted target. Reason: Loss of control/network input must not leave normal actuation enabled.

Acceptance: Stop the source or disconnect its cable: observed laser-off occurs no later than 110 ms after the last fresh accepted target (100 ms timeout plus the existing 10 ms test margin). Invalid packets do not reset it.

Time values: project targets. Timing basis: [C] permanence/action delay; [L] pp. 5, 10.

F-VLT-11 | One-camera omission tolerance

With exactly two valid cameras, continue in indicated degraded mode; with fewer than two, switch the laser off. Reason: One camera loss is the required tolerated node failure.

Acceptance: For each missing-camera case, tracking continues with the other pair; covering/stopping a second camera turns the laser off within 100 ms. Full 80% acceptance in degraded mode remains OQ-14.

Time value: project target. Timing basis: [C] fault hypothesis / membership; [L] p. 4 and pp. 5, 10.

NF-VLT-2 | Laser selection constraint

Use a visible laser meeting the project restriction of at most Class 2; confirm the selected module and local laboratory conditions before laser-on tests. Reason: Laser selection and permitted operating conditions are physical setup constraints.

Acceptance: Record and review the actual module label/datasheet and laboratory authorization. Selection and local rules remain OQ-2/OQ-6; this report is not a certification claim.

NF-VLT-3 | Independent physical interruption

A physical kill switch cuts the laser supply independently of node software. Reason: The operator needs a direct means of removing laser power.

Acceptance: With all nodes running and requesting laser-on, opening the switch removes laser power and the optical output goes off. After release, apply the recovery policy still to be agreed in OQ-11.

Timing, load and non-interference

Lecture references justify what must be bounded and measured. Numerical project targets are identified separately; none is presented as a number prescribed by the slides.

NF-VLT-5 | Shared clock precision

All nodes share a clock with a pairwise offset of at most 1 ms; observations carry exposure time in ns. Synchronization must remain valid during operation. Reason: Different camera observations must refer to comparable physical times.

Acceptance: During a 10 min target run, record every node clock offset and explicitly check the pairwise bound of 1 ms, including maximum monitoring load. Resolve exposure-clock mapping, granularity g and timestamp uncertainty in OQ-12.

1 ms / 10 min: project targets. Basis: [C] global time, g > precision, timestamp uncertainty; [L] pp. 10-11.

NF-VLT-18 | Exposure-to-placement timing

Budget capture/readout, detection, both network transfers, fusion/prediction, output and mirror settling against the project target of 30 ms from exposure to physical spot placement. Reason: Every distributed physical transaction needs an end-to-end timing bound.

Acceptance: Over 10 min on target hardware, report per-stage and total timing, including p99 and maximum. The current draft criterion is p99 <= 30 ms. Use measured maxima in the budget; hard-deadline versus percentile interpretation remains OQ-12.

30 ms / 10 min: project targets. Basis: [L] p. 3 (each transaction), p. 5 (maxima), pp. 10-11 (timestamping/target measurement); [C] deadline.

NF-VLT-6 | Monitoring cannot block the loop

Monitoring is best effort: under back-pressure, discard monitoring data instead of building a backlog or blocking tracking. The system must also work with the laptop absent. Reason: Observation traffic must not change the physical-loop guarantees.

Acceptance: Pause/disconnect monitoring, then run at the NF-VLT-8 maximum: memory remains bounded and NF-VLT-1/NF-VLT-18 (30 ms target) still pass. Record timing with monitoring on and off.

Timing basis: [C] composability / load; [L] pp. 5, 11.

NF-VLT-8 | Monitoring load bound

Aggregate monitoring traffic is at most 20 Mbit/s; camera images are limited to 5 images/s per camera and 320 x 240 px. Reason: Monitoring shares link and compute resources with the loop.

Acceptance: During a 10 min run at maximum configured monitoring, measured traffic respects the bound and image rate/size limits.

5 images/s / 10 min: project targets. Basis: [C] load hypothesis / composability; [L] pp. 5-6 (loaded-network measurement).

NF-VLT-19 | Per-node resource headroom

Each node uses at most 75% average CPU and 50% RAM on the target Pi 3. Reason: An integrated load must not exhaust the node resources used by time-critical work.

Acceptance: Resource logs from a 10 min target run with maximum configured monitoring meet both limits on every node; NF-VLT-18 is checked in the same run.

10 min: project target. Basis: [C] per-node load / freedom from thrashing; [L] pp. 5-6, 11.

What is still deliberately unclaimed

No implementation measurements or completed test results are claimed at M2. The 30 ms, 1 ms and shutdown thresholds are requirements to verify, not evidence of achieved performance. Missing validity limits, measurement granularity and physical settling instrumentation are owned in OQ-12.

DRS checklist coverage

All 13 rows of the exercise DRS checklist [C] are reproduced below. Each has a requirement mapping or an explicit deferral with a reason.

Checklist concern	Requirement or explicit deferral
Global time base	NF-VLT-5 and F-VLT-22: shared exposure time and ongoing sync validity. Granularity g > precision is deferred to OQ-12: the camera clock/mode is not selected. Owner: system design.
Timestamp comparison	NF-VLT-5, F-VLT-3, F-VLT-22. Reasonableness and the checklist uncertainty bound (+/-2g) must be part of the comparison rule. Numerical g is deferred to OQ-12 for the same hardware dependency.
Temporal accuracy / phase sensitivity	F-VLT-4 and F-VLT-22: bound image age, exposure skew and prediction horizon; test boundary and phase cases. Values deferred to OQ-12 pending measured capture/loop timing.
Permanence / action delay / hidden channels	F-VLT-10, F-VLT-22, F-VLT-23, NF-VLT-18. Source time, placement time, expiry and bounded queues must govern use; newly arriving stale data is not permanent/current data. Detailed delay/buffering rule remains OQ-12.
End-to-end transaction deadline	NF-VLT-18: exposure to physical placement, including all processing/transmission/settling stages. Individual budgets are deferred to M3, after Gate 1; measured maxima must be used [L p. 5].
Fault hypothesis / redundancy degree	F-VLT-11: tolerate one camera omission/loss (k = 1), using two of three cameras. F-VLT-8/F-VLT-10/NF-VLT-3 cover laser-off response and physical interruption. Arbitrary-value faults and a frozen galvo are outside the tolerated hypothesis; residual risk is stated in decisions.
Replica determinism / voting	Deferred with optional F-VLT-14: no replicated controller is required for this scope. Reason: spare hardware and takeover scope are undecided (OQ-8); no voting scheme is claimed.
Membership / partitions / degraded mode	F-VLT-11, F-VLT-17, F-VLT-22, F-VLT-23. Camera eligibility follows fresh valid input, including loss, stale packets and restart. Exact recovery/re-arm policy remains OQ-11.
Per-channel/node load / no thrashing	NF-VLT-8, NF-VLT-19, F-VLT-23. Bounded traffic, resource headroom and finite buffering. Functional loop rates and capacities remain OQ-12 because camera mode is open.
Composability	NF-VLT-6: demonstrate the same success/timing requirements with monitoring absent, stalled and at maximum load.
Gateway flow control	Deferred as not applicable: one installation on one wired switch; no gateway between real-time clusters is in scope. Local message buffering is still covered by F-VLT-23.
Monitor actuator effect	F-VLT-2, F-VLT-5, F-VLT-6: observe the physical spot, not only the command. A blocked/misdirected actuator must be visible in measured error/misses; metric validity remains OQ-9.
Node reintegration / minimal history	F-VLT-17, F-VLT-23, F-VLT-24: synchronized fresh session with validated calibration/configuration; reject old history. Exact ground state and rejoin point are deferred to OQ-11 before implementation.

Decisions and their reasons

These are inherited project choices and scope limits. They do not constitute the node architecture or a Gate 1 approval.

Decision	Reason and trace
Five wired Pi 3 nodes	Three camera nodes, one control node and one galvo node are the existing hardware constraint. This meets the exercise minimum of five nodes [L pp. 3-4] and permits a two-of-three camera case (F-VLT-1, F-VLT-3, F-VLT-5, F-VLT-11).
Software clock synchronization	Use the existing software-PTP choice with a measured, rather than assumed, pairwise precision target. No hardware camera trigger in this version (NF-VLT-5, F-VLT-22).
Manual calibration first	Calibrate cameras and galvo manually and retain the result. Automatic galvo calibration is deferred because it depends on dependable spot visibility (F-VLT-3, F-VLT-5, F-VLT-24; OQ-13).
Observe-only monitoring	Monitoring receives pushed data and has no live command authority. This keeps display failure and congestion outside the control dependency; arming remains local (NF-VLT-6, NF-VLT-8; OQ-11/OQ-13).
One-camera-loss tolerance	The required tolerated case is one missing/omitting camera, not arbitrary corrupt data or loss of all computation. Use the remaining pair; fewer than two valid cameras requires laser-off (F-VLT-11, F-VLT-22).
No required redundant controller	Controller redundancy is optional and deferred until hardware availability and takeover scope are settled. Avoid adding replica/voting obligations to the core M2 scope (F-VLT-14 deferred; OQ-8).
Hardware watchdog deferred	The earlier project decision excludes a hardware watchdog. A galvo process frozen with laser-on can retain its last output until the physical kill switch is used; software timeout claims do not cover this hang. This is a residual risk for team review (F-VLT-10, NF-VLT-3).
One laboratory demonstrator	No long-term operation, cloud service, automatic updates or certification claim is included. Local laboratory approval remains necessary before laser-on tests (NF-VLT-2, NF-VLT-3; OQ-6).

Lecture and checklist sources

[L] Distributed Real-Time Systems - Exercise, WS26, 23 October 2026, 12-page slide set supplied with the assignment. Page numbers here are PDF page numbers. Relevant pages: 3-4 system/transaction/fault criteria; 5-6 measurement and maxima; 7 ownership; 8 M2/Gate 1; 9 independent testing; 10 test bench; 11 specification and target measurement; 12 checklist pointer.

[C] Exercise build materials, Requirement Checklists, section "Distributed real-time system (DRS)". All 13 concerns are copied into the checklist coverage table, so this report can be reviewed without opening external project files.

Source rule: a lecture/checklist reference identifies the principle. Numeric deadlines, test durations and frequency limits are project targets unless explicitly stated otherwise. The supplied exercise slides do not prescribe 30 ms, 1 ms or 100 ms.

Open questions and Gate 1

Owners are sub-team roles. A pending item is not recorded as a decision or as implemented behaviour.

ID / owner	Question and affected requirement IDs
OQ-1 **Sensor and actuator modules**	Which camera model, resolution/crop, shutter and frame mode? Measure capture/readout cost; no 90 fps minimum has been established (F-VLT-1, F-VLT-2, NF-VLT-18).
OQ-2 **Sensor and actuator modules**	Which laser module, colour and switching interface meet the project selection constraint and image visibility needs (NF-VLT-2, F-VLT-2, F-VLT-5)?
OQ-3 **System design**	What cube dimensions and starting deflection define the allowed aiming envelope and evaluation setup (F-VLT-9, F-VLT-7)?
OQ-4 / OQ-5 **Sensor and actuator modules**	Confirm galvo signal range, wiring, converter, ability to disable its pattern player and measured settling behaviour (F-VLT-5, F-VLT-9, NF-VLT-18).
OQ-6 **Project management**	Who holds the responsible laboratory role and what local authorization/conditions must be satisfied before laser-on testing? Record roles only (NF-VLT-2, NF-VLT-3).
OQ-7 **Sensor and actuator modules**	Choose background, ball/laser contrast, lighting and exposure for dependable observations (F-VLT-1, F-VLT-2).
OQ-8 **Project management**	Are spare nodes available, and is optional controller redundancy included? Otherwise retain the stated deferral (F-VLT-14 deferred).
OQ-9 **Requirements and product owner**	Define aiming-error geometry in mm, single-camera spot validity, background false hits, unmeasurable cases and independent moving-position/error references (F-VLT-4, F-VLT-6, F-VLT-7, NF-VLT-1).
OQ-10 **Requirements and product owner**	Define actual release, amplitude window/end, swing-period duration in s, time weighting/cadence, gaps and aborted runs. Missing fusion output must not vanish from scoring (F-VLT-7, NF-VLT-1).
OQ-11 **System design**	Unify restart, transient loss, timeout, sync loss, alarm and kill-switch recovery in button/auto modes. Define switch feedback, ground state, retained state and rejoin/re-arm point (F-VLT-8, F-VLT-10, F-VLT-11, F-VLT-17, NF-VLT-3).
OQ-12 **System design**	Set age/skew/horizon/sync-status and queue limits; define granularity/uncertainty, pairwise clock checks and physical placement measurement. Resolve hard 30 ms deadline versus current p99 criterion (F-VLT-22, F-VLT-23, NF-VLT-5, NF-VLT-18).
OQ-13 **Integration support**	Fix units, axes/handedness, transforms, image convention, setup revision and manual calibration/test-point procedures compatible with local arming (F-VLT-3, F-VLT-5, F-VLT-24).
OQ-14 **Requirements and product owner**	Agree operating envelope, combined spatial/timing error budget and whether every degraded camera pair must also meet the 80% acceptance (F-VLT-1 to F-VLT-5, F-VLT-11, NF-VLT-1).